

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

LOW-RANK MATRIX FACTORIZATION

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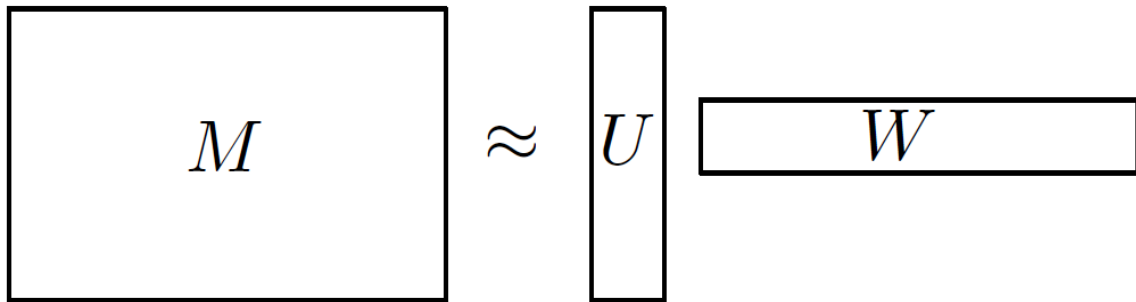


Low-rank Matrix Approximation

Let M be any $p \times q$ matrix.

Want to approximate M by a $p \times q$ matrix $\hat{M}$ of the form UW :

- U is $p \times k$ and W is $k \times q$
- $k \leq p, q$ is of our choosing


$$M \approx U W$$

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- 1 Compact representation
- 2 Method of embedding data

Generalizing the Spectral Decomposition

For **symmetric** matrices (e.g., covariance matrices), we have seen:

- Results about existence of eigenvalues and eigenvectors
- Eigenvectors form an alternative basis
- Resulting spectral decomposition, used in PCA

What about **arbitrary** matrices $M \in \mathbb{R}^{p \times q}$?

Singular Value Decomposition (SVD)

Any $p \times q$ matrix ($p \leq q$) has a **singular value decomposition**:

$$M = \underbrace{\begin{pmatrix} \updownarrow u_1 & \cdots & \updownarrow u_p \end{pmatrix}}_{p \times p \text{ matrix } U} \underbrace{\begin{pmatrix} \sigma_1 & \cdots & 0 \\ \cdots & \ddots & \cdots \\ 0 & \cdots & \sigma_p \end{pmatrix}}_{p \times p \text{ matrix } \Lambda} \underbrace{\begin{pmatrix} \leftarrow v_1 \rightarrow \\ \leftarrow \vdots \rightarrow \\ \leftarrow v_p \rightarrow \end{pmatrix}}_{p \times q \text{ matrix } V^T}$$

- $u_1, \dots, u_p$ are orthonormal vectors in $\mathbb{R}^p$
- $v_1, \dots, v_p$ are orthonormal vectors in $\mathbb{R}^q$
- $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0$ are **singular values**



Concisely Approximating a Matrix

Singular value decomposition of $p \times q$ matrix M (with $p \leq q$):

$$M = \begin{pmatrix} \updownarrow u_1 & & \\ & \cdots & \\ & & \updownarrow u_p \end{pmatrix} \begin{pmatrix} \sigma_1 & \cdots & 0 \\ \cdots & \ddots & \cdots \\ 0 & \cdots & \sigma_p \end{pmatrix} \begin{pmatrix} \leftarrow v_1 \rightarrow \\ \vdots \\ \leftarrow v_p \rightarrow \end{pmatrix}$$

A concise approximation to M , for any $k \leq p$

$$\hat{M} = \underbrace{\begin{pmatrix} \updownarrow u_1 & & \\ & \cdots & \\ & & \updownarrow u_k \end{pmatrix}}_{p \times k} \underbrace{\begin{pmatrix} \sigma_1 & \cdots & 0 \\ \cdots & \ddots & \cdots \\ 0 & \cdots & \sigma_k \end{pmatrix}}_{k \times k} \underbrace{\begin{pmatrix} \leftarrow v_1 \rightarrow \\ \vdots \\ \leftarrow v_k \rightarrow \end{pmatrix}}_{k \times q}$$

Optimality Property: Best Low-rank Approximation

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SVD yields the best such approximation $\hat{M}$, minimizing the squared error

$$\sum_{i,j} (M_{ij} - \hat{M}_{ij})^2.$$